HCL-Net: Heterogeneous Collaborative Learning for Lightweight Remote Sensing Image Segmentation

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***Abstract*—Semantic segmentation of remote sensing images remains challenging due to large intraclass variations, high interclass similarity, and the demand for lightweight deployment. Conventional single-architecture models and homogeneous collaborative frameworks struggle to balance local detail extraction with global context modeling. To address these limitations, we propose heterogeneous collaborative learning (HCL-Net), a HCL-Net framework that integrates convolutional and Transformer architectures. HCL-Net consists of two complementary student networks: the frequency-domain local detail network (FLDNet), based on ResNet18 with a wavelet phase–amplitude fusion block (WPAFB) to capture multifrequency information, and the spatial-domain global structure network (SGSNet), built on a DFormer-T backbone with a dynamic texture–edge perception (DTEP) module for robust global context modeling. A dual collaborative strategy enhances knowledge transfer between networks through: 1) bidirectional feature reconstruction (BFR), which aligns high-order statistics using Gram matrix alignment and enforces feature-space consistency via variational information distillation (VID); and 2) regional pixel-level contrastive learning, which improves intraclass compactness while reducing interclass confusion. Experiments on the Vaihingen dataset demonstrate that collaborative training yields substantial gains over independent training, with FLDNet achieving mAcc 89.92%/mean intersection over union (mIoU) 82.12% and SGSNet achieving mAcc 89.95%/mIoU 82.16%, improving accuracy by 2.26%/2.08% and IoU by 2.53%/2.19%. With only 24.25 and 12.36 M parameters and computational costs of 6.13 and 6.35 G, FLDNet and SGSNet outperform 19 state-of-the-art (SOTA) methods while remaining efficient for resource-constrained environments. The code and experimental results are available at https://github.com/110-011/HCL-Net**

***Index Terms*—Dynamic texture perception, heterogeneous collaborative learning (HCL-Net), pixel contrastive learning, semantic segmentation, wavelet phase–amplitude fusion.**

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I. Introduction

With the rapid development of satellite remote sensing and image processing technologies, the semantic segmentation of remote sensing images has emerged as a key technology having important applications in the fields of urban planning [1], disaster assessment [2], and environmental monitoring [3]. However, remote sensing images typically possess complex-object distributions coupled with large intraclass variations and high interclass similarities, which pose significant challenges for existing semantic segmentation algorithms. Furthermore, in practical applications, models must typically be deployed on resource-constrained mobile or portable devices, necessitating higher computational efficiency and increasingly lightweight models.

Over the past decade, convolutional neural network (CNN)-based models [4], [5] have advanced semantic segmentation by enabling the extraction of local features. However, CNNs exhibit spatial invariance, which limits their use for modeling contextual relationships among pixels. To overcome this problem, researchers have employed dilated convolutions [6], pyramid networks [7], and spatial–channel attention [8], along with the stacking of multiple convolutional blocks and network deepening [9], [10], [11] for enhanced context modeling, though accuracy bottlenecks remain when identifying densely packed small targets and objects with complex textures. Recently, breakthroughs have been achieved for Transformer architectures (e.g., the Swin Transformer [12]) regarding the understanding of complex scenes, with self-attention mechanisms being used to capture long-range dependencies. However, these architectures have high computational complexity and strict hardware requirements, which limit their applicability on edge devices. Recent advances have further addressed specific challenges through interactive representation refinement approaches that combine geometric priors to reduce semantic ambiguity while maintaining local details [13] and global adaptive second-order Transformers that enhance global context capture by addressing conventional Vision Transformers' limitations [14]. Despite these developments, the key problem remains that no single architecture can simultaneously balance global context modeling and local detail extraction, highlighting the ongoing need for architectures that effectively integrate global context understanding with local detail preservation.

In recent years, knowledge distillation (KD) and collaborative learning (CL) have provided new approaches to collaborative multimodel optimization. Traditional KD transfers

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knowledge from a large pretrained teacher model to a lightweight student network [15], [16], [17], [18], [19], [20], whereas CL promotes the joint evolution of equally sized models through bidirectional knowledge sharing [21], [22], [23]. However, most existing methods are based on knowledge transfer among homogeneous models, the similar structures of which yield a singular feature perspective. Heterogeneous models have greater potential because of their complementary features; however, designing an effective collaborative mechanism to fully exploit the advantages of these models and achieve efficient lightweight knowledge transfer remains a key challenge.

To address the challenges outlined above, this study proposes a heterogeneous CL framework heterogeneous collaborative learning (HCL-Net) based on complementary feature reconstruction and regional pixel contrastive learning (RPCL). The framework comprises two complementary heterogeneous student models that utilize wavelet phase–amplitude fusion block (WPAFB) and dynamic texture–edge perception (DTEP) module for frequency- and spatial-domain feature fusion, respectively. Additionally, bidirectional feature reconstruction (BFR) and RPCL strategies are introduced to achieve effective cross-model complementary knowledge transfer and feature alignment.

The main contributions of this study are as follows.

1) *Complementary Heterogeneous Dual-Model Design:* A dual-model architecture with complementary functions is proposed. The frequency-domain local detail network (FLDNet), which is the first student model, combines ResNet18 [24] with a WPAFB module to separate low- and high-frequency components (for illumination invariance and detail preservation, respectively) via multiscale wavelet decomposition; hence, the robustness to illumination interference and detail retention are both optimized. The spatial-domain global structure network (SGSNet), the second student model, is based on a DFormer-T [25] backbone and utilizes a DTEP module to dynamically fuse multidirectional texture features and structural edge information. Together, the FLDNet and SGSNet models form a complementary system for frequency–spatial expression and local detail to global context modeling to effectively address the challenges of complex-object recognition.
2) *Bidirectional CL Mechanism:* An innovative bidirectional CL mechanism is proposed, in which BFR aligns the high-order feature statistical distributions via Gram matrix alignment and constrains the feature-space consistency using variational information distillation (VID) [26]; this approach effectively mitigates domain discrepancies between the heterogeneous models. Furthermore, RPCL is used to construct pixel-level similarity constraints at the feature level, thereby enhancing the intraclass compactness and suppressing confusion. This approach addresses the limitations in existing methods regarding the handling of objects with large intraclass variations and high interclass similarities.
3) *Efficient Lightweight Implementation:* The proposed framework achieves high accuracy (mAcc 89.92%/89.95% and mean intersection over union (mIoU) 82.12%/82.16% on the Vaihingen dataset) while maintaining low parameter counts (24.25 and 12.36 M for FLDNet and SGSNet, respectively) and low computational costs (6.13 and 6.35 G, respectively), thereby enabling deployment in resource-constrained environments.

II. Related Work

A. Remote Sensing Image Semantic Segmentation

Remote sensing semantic segmentation assigns each pixel in aerial or satellite imagery to a land-cover category. With the rise of deep learning, Volpi and Tuia [27] pioneered the use of fully convolutional networks (FCNs) for high-resolution segmentation, achieving substantial gains through end-to-end training. Building on this, Ronneberger et al. [5] proposed the U-Net architecture, which uses encoder–decoder pathways with skip connections to recover spatial detail. U-Net and its variants have since become standard in remote sensing, mitigating feature loss during propagation. To expand receptive fields, Chen et al. [28] introduced dilated convolutions and atrous spatial pyramid pooling, significantly improving urban object classification. Other advances include multiscale fusion networks with residual and attention modules [29] and boundary-aware architectures targeting precise edge delineation [30]. However, CNN-based models remain limited by local receptive fields, hindering performance in complex scenes that demand global context.

Inspired by the success of Transformers, Liu et al. [12] proposed the Swin Transformer, capturing long-range dependencies via hierarchical, window-based attention. SegFormer [31] combined a lightweight Transformer encoder with a multilevel decoder to balance accuracy and efficiency, while Jiang et al. [32] introduced a graph Transformer leveraging adaptive attention for complex pixel relationships. Hybrid approaches have also emerged, such as multimodal CNN–Transformer fusion [33] and cascaded networks integrating atrous convolutions, spatial pyramids, and Swin modules [34]. Beyond RGB, multimodal inputs [thermal, depth, and normalized digital surface model (nDSM)] further enrich segmentation. Dual-branch fusion frameworks [35], [36], [37], [38], [39], [40], [41], which process modalities separately and integrate them hierarchically, have proven especially effective.

Despite these advances, remote sensing images still pose challenges due to intraclass variation, interclass similarity, and illumination changes. CNNs capture local textures but lack global awareness; the Transformers model global relations but often miss edge details and impose high computational costs. These complementary strengths and weaknesses underscore the need for heterogeneous collaborative models that combine CNN and Transformer capabilities within lightweight frameworks.

B. Collaborative Learning

CL enables multiple models to evolve through mutual optimization and knowledge exchange. Early efforts in KD,

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introduced by Hinton et al. [15], transferred information from a large teacher to a smaller student, later enhanced with intermediate feature cues [16]. For segmentation, structured KD [42] transferred pixel relationships to improve boundary accuracy, while channel-wise [43] and cross-image relational KD [44] targeted foreground alignment and relational consistency. However, these methods rely on pretrained teachers and unidirectional transfer, constraining student potential.

CL addresses these limitations by promoting bidirectional interaction. Zhang et al. [21] pioneered deep mutual learning, encouraging coevolution through output distribution alignment. Wu et al. [45] extended this with online distillation across multibranch networks, while Wu and Gong [46] introduced peer learning to mitigate noisy labels via reciprocal filtering. Guo et al. [47] further advanced feature interaction through attention-guided exchange.

Although CL has shown strong promise in general vision tasks [21], [45], [46], [47], its application in remote sensing remains underexplored. Most existing studies rely on KD [20], [42], [43], [44] or train models independently. Moreover, current CL frameworks predominantly involve homogeneous models, limiting feature diversity and failing to capture the complex, multimodal, and multiscale characteristics of remote sensing data. Heterogeneous collaboration is particularly challenging, as mismatched feature spaces across architectures cannot be reconciled by simple distribution alignment. To bridge this gap, we propose a BFR strategy that aligns high-order statistics via Gram matrix matching and enforces distributional consistency through VID, enabling effective collaboration between heterogeneous models.

C. Contrastive Learning

Contrastive learning improves feature discrimination by drawing semantically similar samples closer and pushing dissimilar ones apart. Recent advances include MoCo [48], which introduced momentum encoders and dynamic dictionaries; SimCLR [49], which highlighted the role of augmentations and projection heads; and BYOL [50], which achieved self-distillation without negative samples. For dense prediction, Wang et al. [51] extended contrast to pixel-level features, Xie et al. [52] designed pixel-level pretext tasks, and Zhou and Wang [53] incorporated global context to capture structural pixel relationships.

Yet, applying contrastive learning to remote sensing segmentation presents three issues. First, many approaches remain at the image or region level [48], [49], [51], lacking fine-grained pixel discrimination for complex objects. Second, pixel-level methods [52], [53] often ignore severe class imbalance, allowing dominant classes to overshadow rare ones. Third, most frameworks are single models, underutilizing the potential of heterogeneous collaboration.

To address these gaps, we introduce an RPCL strategy tailored to heterogeneous models. RPCL employs region-aware hierarchical sampling to balance class representation, adaptive temperature scaling for stable optimization, and multiscale collaborative contrast to leverage complementary architectures. This represents a step toward fully exploiting heterogeneous collaboration for robust pixel-level representation in remote sensing.

III. Proposed Method

A. Overview

The proposed HCL-Net framework comprises FLDNet and SGSNet, two heterogeneous student models with complementary structures and feature representations, along with a bidirectional CL mechanism. The framework is designed to address three major challenges arising in the semantic segmentation of remote sensing images: 1) large intraclass variations (e.g., for buildings under different lighting conditions); 2) high interclass similarities (e.g., for low vegetation versus trees); and 3) the need for lightweight models suitable for deployment on edge devices. The overall HCL-Net workflow is as follows. First, the two heterogeneous and complementary models are built to capture comprehensive information from the remote sensing images. FLDNet employs a ResNet18 backbone combined with a WPAFB module to enable illumination robustness and detail preservation (Fig. 1). SGSNet incorporates a DFormer-T backbone with a DTEP module to enhance the global context and structural information (Fig. 2). Together, these models form a complementary system for frequency–spatial expression and local detail to global context modeling that effectively addresses complex-object recognition challenges. Second, during training, a dual CL mechanism (Fig. 3) is designed, which employs BFR to align high-order feature statistics and RPCL to enhance intraclass feature compactness. Hence, complementary knowledge transfer between the heterogeneous models is achieved.

B. Complementary Heterogeneous Model Design

FLDNet and SGSNet are complementary in structure and feature expression.

1) FLDNet: This model incorporates a dual-channel encoder based on ResNet18 to extract multiscale features for both RGB and nDSM modalities. The WPAFB module performs the adaptive weighted fusion of low- and high-frequency information (for illumination robustness and detail preservation, respectively) in the frequency domain. Hence, a balanced representation of the local details and global illumination information is obtained. FLDNet is a multiscale dual-branch feature fusion model. Two independent ResNet18 encoders (one each for the RGB and nDSM branches) extract multiscale features, yielding R_1–R_5 features for RGB and N_1–N_5 features for nDSM. For each scale pair (R_i, N_i), the WPAFB module fuses the features to obtain the fused feature $f_{1,i}$ (where $i \in \{1, 2, 3, 4, 5\}$). Finally, a cascaded decoder module performs progressive upsampling and skip-connection fusion of the multiscale fused features to generate the output out_1. The WPAFB structure (indicated in orange in Fig. 1) is described in the following.

WPAFB: First, a discrete wavelet transform [DWT(·)] is applied to the input features X_R (N_i) and X_R (N_i) to decompose them into the following low- and high-frequency components:

$$(L_R, H_R) = \text{DWT}(X_R) \tag{1}$$

$$(L_N, H_N) = \text{DWT}(X_N). \tag{2}$$

Then, taking the low-frequency features L_R and L_N as an example, a fast Fourier transform [FFT(·)] is used to extract the phase ϕ and amplitude A of each component. Two learnable

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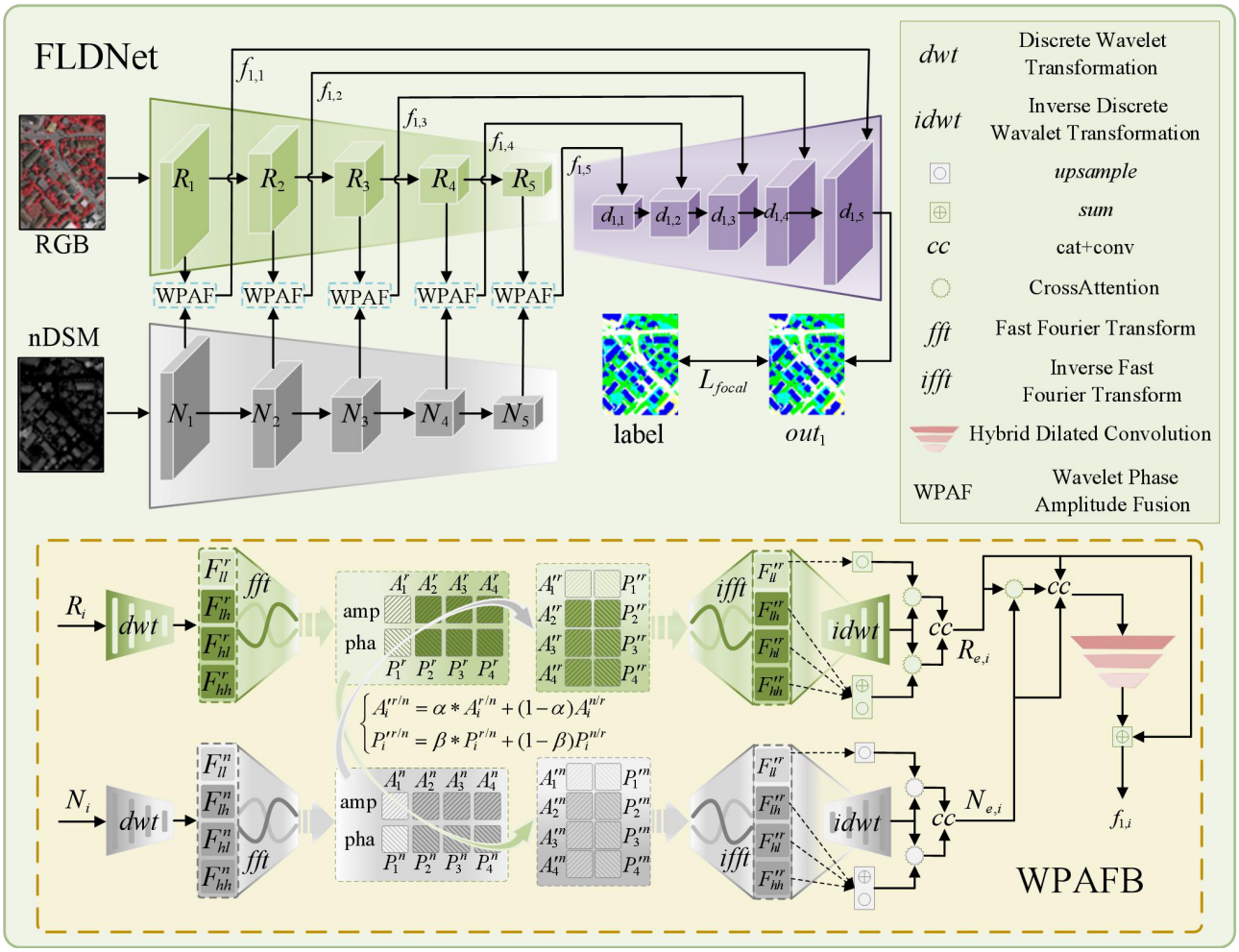

Fig. 1. Overview of FLDNet architecture.

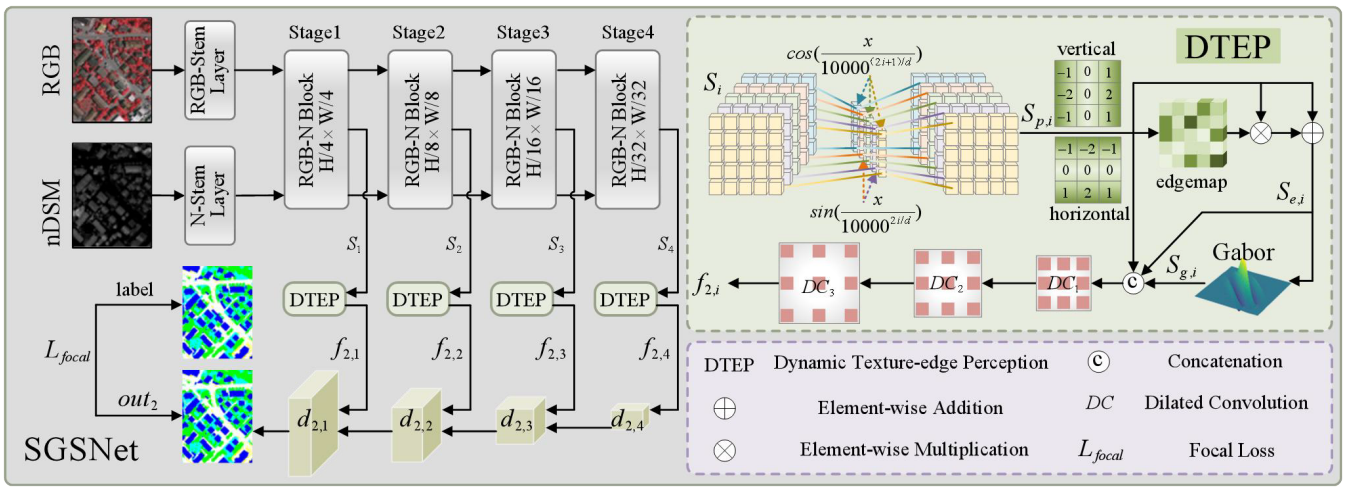

Fig. 2. Overview of SGSNet architecture.

parameters α and β (mapped via a Sigmoid function to the interval [0,1]) are introduced to enable adaptive weighted fusion, which yields the fused phase ϕ_{fused} and amplitude A_{fused}; then, an inverse fast Fourier transform [IFFT(·)] is applied sequentially to ϕ_{fused} and A_{fused} to reconstruct the following low-frequency components:

$$F_R = \text{FFT}(L_R), \phi_R = \angle(F_R), A_R = |F_R| \tag{3}$$

$$F_N = \text{FFT}(L_N), \phi_N = \angle(F_N), A_N = |F_N| \tag{4}$$

$$\phi_{\text{fused},R} = \sigma(\alpha) \cdot \phi_R + (1 - \sigma(\alpha)) \cdot \phi_N \tag{5}$$

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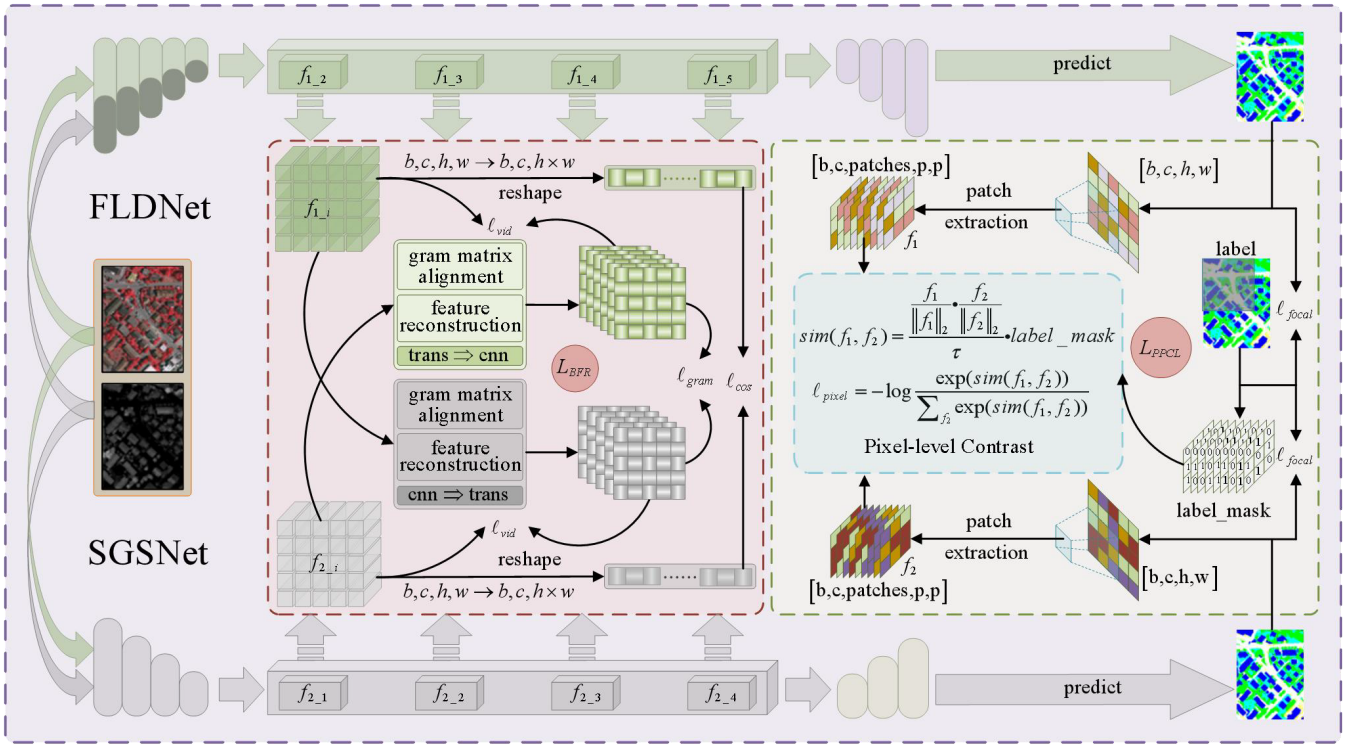

Fig. 3. Overview of HCL-Net architecture.

$$A_{\text{fused},R} = \sigma(\beta)\cdot A_R + (1-\sigma(\beta))\cdot A_N \tag{6}$$

$$\phi_{\text{fused},N} = \sigma(\alpha)\cdot \phi_N + (1-\sigma(\alpha))\cdot \phi_R \tag{7}$$

$$A_{\text{fused},N} = \sigma(\beta)\cdot A_N + (1-\sigma(\beta))\cdot A_R \tag{8}$$

$$L_{\text{fuse},R} = \text{IFFT}\left(A_{\text{fuse},R}\cdot e^{i\phi_{\text{fused},R}}\right) \tag{9}$$

$$L_{\text{fused},N} = \text{IFFT}\left(A_{\text{fused},N}\cdot e^{i\phi_{\text{fused},N}}\right) \tag{10}$$

where $\sigma(\cdot)$ represents the Sigmoid function. For the high-frequency components, a similar fusion process is applied to each corresponding wavelet subband to obtain $H_{\text{fused},R}$ and $H_{\text{fused},N}$.

Next, to further fuse the cross-modal features, the reconstructed components are transformed back to the spatial domain using the inverse discrete wavelet transform [$\text{IDWT}_{(\cdot)}$]; hence, the reconstructed features F_R and F_N are obtained. Self-attention is then applied to F_R and F_N and combined with upsampled low- ($L_{R,\text{up}}$ and $L_{N,\text{up}}$) and high-frequency ($H_{R,\text{up}}$ and $H_{N,\text{up}}$) information (matched via bilinear interpolation). As a result, the enhanced features E_R and E_N are obtained. Finally, a cross-attention mechanism is used to fuse E_R and E_N to produce the fused feature C. To preserve the explicit spatial feature information, C is concatenated with E_R and E_N. For long-range feature extraction, the result is then processed via convolution and a hybrid dilated convolution [$\text{HDC}_{(\cdot)}$] module (comprising dilated convolutions with different dilation rates of 1, 2, and 3). The final fused feature f is then obtained. Thus,

$$F_R = \text{IDWT}\left(L_{\text{fused},R}, H_{\text{fused},R}\right) + X_R \tag{11}$$

$$F_N = \text{IDWT}\left(L_{\text{fused},N}, H_{\text{fused},N}\right) + X_N \tag{12}$$

$$E_R = \text{EnhanceAttention}\left(F_R, \left[L_{R,\text{up}}, H_{R,\text{up}}\right]\right) \tag{13}$$

$$E_N = \text{EnhanceAttention}\left(F_N, \left[L_{N,\text{up}}, H_{N,\text{up}}\right]\right) \tag{14}$$

$$C = \text{CrossAttention}\left(E_R, E_N\right) \tag{15}$$

$$f = \text{HDC}\left(\text{ConvCat}\left([C, E_R, E_N]\right)\right) + F_R. \tag{16}$$

Finally, for each input scale (R_i, N_i), the output fusion features of the WPAFB module are as follows:

$$f_{1,i} = \text{WPAFB}_i\left(R_i, N_i\right) \tag{17}$$

2) SGSNet: This model employs a pretrained DFormer-T encoder to extract multiscale features and incorporates a DTEP module to enhance the positional information, edge features, and texture details at different scales. This design compensates for the inadequacy of the Transformer architecture in capturing fine-grained local features while supplementing the global context information with edge and texture information; hence, a balance is achieved between local and global representations. SGSNet is a single-stream model that uses a DFormer-T encoder to extract four different scale features (S_1–S_4) from the input image (with the multimodal fusion being handled by the encoder). For each scale, the DTEP module performs detail enhancement and information fusion to obtain the enhanced feature $f_{2,i}$ (where $i \in \{1, 2, 3, 4\}$). Finally, a cascaded decoder module upsamples and fuses the multiscale features via skip connections to generate the output out_2. The DTEP module (indicated in green in Fig. 2) is formulated as follows.

a) DTEP: To inject explicit positional information for each spatial location globally, the input feature S_i is first augmented with positional encoding. This process consists of two parts: 1) sine and cosine encodings are added to even and odd channels, respectively; and 2) an independent learnable

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bias θ for each channel is introduced at every spatial location. The position-encoded feature is denoted by $S_{p,i}$,i.e.,

$$\mathrm{PE}(x, y, 2k) = \sin\left(x \cdot 10000^{-2k/d}\right) \tag{18}$$

$$\mathrm{PE}(x, y, 2k+1) = \cos\left(x \cdot 10000^{-2k+1/d}\right) \tag{19}$$

$$S_{p,i} = S_i + \mathrm{PE} + \theta \tag{20}$$

where (x, y) denotes the spatial position on the feature graph, k represents the channel index, d represents the total channel number, and θ is the independently learned parameter matrix.

Next, to extract the edge structural information, the Sobel operator is applied to compute the gradients of feature map $S_{p,i}$ in the x- and y-directions. These gradients are then combined to obtain the gradient magnitude G_i. Then, a 1×1 convolution map G_i for the target number of channels is used. A Sigmoid function is applied to normalize G_i to an edge attention map A_{edge}. Finally, A_{edge} is used to augment $S_{p,i}$ via residual concatenation such that the edge-augmented features $S_{e,i}$ are obtained. The formulas are as follows:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \tag{21}$$

$$G_i = \sqrt{\left(G_x * S_{p,i}\right)^2 + \left(G_y * S_{p,i}\right)^2} \tag{22}$$

$$A_{\mathrm{edge}} = \sigma\left(\mathrm{Conv}_{1\times1}(G_i)\right) \tag{23}$$

$$S_{e,i} = S_{p,i} \cdot A_{\mathrm{edge}} + S_{p,i}. \tag{24}$$

Subsequently, for $S_{e,i}$, a Gabor filter is applied to extract multidirectional (0°, 45°, 90°, and 135°) texture features. All directional texture features are then spliced and integrated via 1×1 convolution to obtain the texture-enhanced feature $S_{g,i}$. Finally, $S_{p,i}$, $S_{e,i}$, and $S_{g,i}$ are spliced and fused by HDC(·) to generate the following final enhanced features $f_{2,i}$:

$$\mathrm{Gabor}(x, y, \lambda, \theta, \sigma, \gamma) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right)\cos\left(\frac{2\pi}{\lambda}x'\right) \tag{25}$$

$$S_{g,i} = \mathrm{ConvCat}\left(\mathrm{Gabor}_j\left(S_{e,i}\right)\right),\ j = 1, 2, 3, 4 \tag{26}$$

$$f_{2,i} = \mathrm{HDC}\left(\mathrm{ConvCat}\left(S_{p,i}, S_{e,i}, S_{g,i}\right)\right) \tag{27}$$

where $x' = x\cos q + y\sin q$; $y' = -x\sin q + y\cos q$; and λ, θ, σ, and γ denote the wavelength, orientation, scale, and spatial aspect ratio, respectively.

b) Cascaded decoder and output generation: In the decoding stage, both models employ a U-Net-type cascade decoder structure that gradually fuses the low- and high-resolution information using jump connections and upsamples the fused features back to the original image size. For Model 1, each layer of feature $f_{1,i}$ is decoded by the decoder block, and the multiscale information is then fused by a residual connection to yield the decoded feature d_i. This process is repeated until the feature map is restored to the original size and the final output out_1 is obtained. The decoder block incorporates operations such as 1×1 dimensionality reduction, deconvolution-based upsampling, and 1×1 convolution to restore the channel size. The SGSNet decoding process is similar and is formulated as follows:

$$\begin{cases} d_i = \mathrm{DB}\left(f_{1,i}\right) + f_{1,i-1}, i = 5, 4, 3, 2 \\ \mathrm{out}_1 = \mathrm{DB}\left(d_2\right) \end{cases} \tag{28}$$

where $\mathrm{DB}(\cdot)$ indicates the decoder block and out_2 denotes the output.

C. Bidirectional CL Mechanism

Because FLDNet and SGSNet focus on frequency-domain local and spatial-domain global modeling, respectively, the features they extract differ significantly in scale, statistical distribution, and semantic expression. Direct fusion or simple transfer cannot overcome domain discrepancies and yield incomplete knowledge transfer. Therefore, a dual CL strategy was designed in this study. In this approach, BFR and RPCL are employed to fully exploit the complementary advantages of the heterogeneous models and achieve precise feature-space alignment and enhanced discriminability. The structure of the bidirectional CL strategy is shown in Fig. 3 and is implemented as described in the following.

1) Bidirectional Feature Reconstruction: BFR is a knowledge transfer mechanism designed to promote complementary knowledge exchange between FLDNet and SGSNet. Its core functions are to perform feature-space mapping and align the statistical properties such that the two models can learn mutually, while their advantages are preserved. The BFR mechanism has three key components: 1) Gram matrix alignment, which captures and transfers high-order feature statistics; 2) a cosine similarity consistency constraint that preserves the overall structure of the feature space; and 3) VID that constrains the distribution of the mapped features. The BFR process (highlighted in pink in Fig. 3) is described in detail in the following.

First, to bridge the feature-space differences between FLDNet and SGSNet, we design lightweight reconstruction modules $R_{1\to2}$ and $R_{2\to1}$ for corresponding layer features F_1 and F_2. Each reconstruction module comprises three sequential components: 1) a 1×1 convolutional layer for channel dimension alignment; 2) batch normalization for feature stabilization; and 3) ReLU activation for nonlinearity. Specifically, when mapping from FLDNet to SGSNet feature space, $R_{1\to2}$ takes input features $F_1 \in \mathbb{R}^{B\times C1\times H\times W}$ and outputs reconstructed features $F_{1\to2} \in \mathbb{R}^{B\times C2\times H\times W}$, where $C1$ and $C2$ represent the channel dimensions of FLDNet and SGSNet features, respectively. The reconstruction process is formulated as

$$F_{1\to2} = \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times1}(F_1)\right)\right) \tag{29}$$

where $\mathrm{Conv}_{1\times1}(\cdot)$ represents the 1×1 convolution operation that maps $C1$ channels to $C2$ channels, $\mathrm{BN}(\cdot)$ denotes batch normalization, and $\mathrm{ReLU}(\cdot)$ is the rectified linear unit activation.

For alignment, we employ Gram matrices rather than direct feature comparison (e.g., Frobenius norm) because Gram matrices $G(F) = F^{\mathrm{T}} \cdot F$ capture second-order feature statistics—specifically, the correlations between different feature channels. This approach focuses on style and texture

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relationships rather than spatial content, making it more robust to architectural differences between heterogeneous models. While the Frobenius norm would enforce pixel-wise similarity, Gram matrix alignment allows features to maintain different spatial patterns while sharing similar statistical distributions, which is crucial for effective knowledge transfer between CNN-based FLDNet and Transformer-based SGSNet architectures

$$G(F) = \frac{1}{BCHW} \cdot F_{BC\times HW} \cdot F_{BC\times HW}^{\mathrm{T}} \tag{30}$$

$$L_{\mathrm{gram},1\to 2} = L_{\mathrm{mse}} = \|G(F_{1\to 2}) - G(F_2)\|_2^2 \tag{31}$$

where $G(\cdot)$ represents the Gram matrix calculation and $F_{BC\times HW}$ is the matrix that reshapes the feature map into batch $\times$ channel $\times$ height $\times$ width dimensions.

Subsequently, to ensure semantic consistency between the two feature groups, the directional similarity of the mapped features is constrained via the following cosine similarity loss $L_{\mathrm{cos},1\to 2}$:

$$L_{\mathrm{cos},1\to 2} = 1 - \frac{F_{1\to 2} \cdot F_2}{\|F_{1\to 2}\| \cdot \|F_2\|}. \tag{32}$$

Then, the VID loss $L_{\mathrm{vid},1\to 2}$ is introduced to model the uncertainty through the predicted feature variance. As a result, the mapped-feature distribution is more comprehensively constrained

$$L_{\mathrm{vid},1\to 2} = \frac{1}{2} \cdot \left(\log\left(\sigma^2\right) + \frac{(F_{1\to 2} - F_2)^2}{\sigma^2} \right) \tag{33}$$

where σ is a learnable variance parameter, $\sigma^2 = \log(1 + e^{\alpha}) + \varepsilon, \alpha$ is the model parameter, and ε is a minimum value.

Finally, the above losses are integrated as follows:

$$L_{\mathrm{BFR},1\to 2} = L_{\mathrm{gram},1\to 2} + L_{\mathrm{cos},1\to 2} + L_{\mathrm{vid},1\to 2}. \tag{34}$$

The overall reconstruction loss is then integrated to yield a bidirectional reconstruction loss. The same process is then applied from SGSNet to FLDNet, yielding $L_{\mathrm{BFR},2\to 1}$.

2) Regional Pixel Contrastive Learning: RPCL is designed to improve feature discriminability in remote sensing image segmentation by enforcing intraclass compactness while maximizing interclass separation. Unlike traditional global contrastive learning or per-pixel contrastive learning, RPCL operates within local patches while incorporating global contrast to achieve a more comprehensive optimization of feature representations. Its core innovations are the stratified sampling strategy, adaptive temperature regulation mechanism, and global–local synergistic comparison framework. The process of RPCL (illustrated in gray in Fig. 3) consists of several components, described below.

a) Local pixel contrastive learning: The goal of this strategy is to enhance local semantic coherence by dividing the feature map into fixed-size patches via a sliding window and constructing pixel-level contrast learning objectives within each patch. The specific steps are as follows.

b) Patch extraction: Given the output feature maps F_1 and $F_2 \in \mathbb{R}^{B\times C\times H\times W}$ from two heterogeneous models, we first use the unfold$(\cdot)$ operation to partition the feature map into a graph of size $K \times K$ of nonoverlapping patches as

$$F = \mathrm{unfold}(F_i, K). \tag{35}$$

This operation converts the feature map into a new shape $F \in \mathbb{R}^{B\times C\times K^2\times N}$, where N is the total number of blocks. For the label Y, the same unfold operation is applied to obtain the patch label y as

$$y = \mathrm{unfold}(Y, K). \tag{36}$$

c) Stratified pixel sampling: Since the category distribution of remote sensing images is usually uneven, direct uniform sampling may result in a few categories of pixels being ignored. We use stratified pixel sampling to ensure that each category of pixels is sampled sufficiently. First, for each category c, we calculate its pixel share r_c in the current block. We then determine the number of samples s_c for each category based on r_c, and we finally randomly sample s_c indexes of pixels for each category to ensure that rare categories are not ignored as follows:

$$r_c = \frac{N_c}{K^2} \tag{37}$$

$$s_c = \tau \cdot r_c \cdot N_{\mathrm{samples}} \tag{38}$$

where N_c is the number of pixels of category c, τ is the equalization factor, and N_{samples} is the total number of samples.

d) Pixel contrastive loss: For each patch, we define the set of positive samples $P(i) = \{j | y_j = y_i\}$ (pairs of pixels of the same category) and the set of negative samples $N(i) = \{j | y_j \neq y_i\}$ (pairs of pixels of different categories), with the contrastive loss defined as follows:

$$\mathrm{sim}\left(f_i, f_j\right) = \frac{f_i f_j}{\|f_i\|\|f_j\|} \tag{39}$$

$$L_i = -\log \frac{\sum_{p\in P(i)} \exp\left(\frac{\mathrm{sim}(f_i, f_p)}{\tau}\right)}{\sum_{p\in P(i)} \exp\left(\frac{\mathrm{sim}(f_i, f_p)}{\tau}\right) + \sum_{n\in N(i)} \exp\left(\frac{\mathrm{sim}(f_i, f_n)}{\tau}\right)} \tag{40}$$

where $\mathrm{sim}(f_i, f_j)$ denotes the cosine similarity between f_i and f_j, and τ is a temperature parameter updated via an adaptive strategy. The contrastive loss for the entire patches is

$$L_{\mathrm{patch}} = \frac{1}{N} \sum_{i=1}^{N} L_i \tag{41}$$

where N represents the number of patches in the current batch.

e) Global semantic contrastive learning: The goal of this strategy is to enhance the global semantic consistency of the model by making the global features of the same image closer together in the feature space while maintaining a greater distance between the global features of different images. Given feature maps F_1 and F_2 from two heterogeneous models (FDFNet and SGSNet), we first apply a global projection head, which is used to map high-dimensional features to a more compact representation space, to obtain $F_{\mathrm{global},1}$ and $F_{\mathrm{global},2}$. This projection head consists of two 1×1 convolutional layers and a ReLU activation function as follows:

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$$F_{\text{global},i} = \text{GlobalProj}\left(F_i\right) = \text{ReLU}\left(\text{Conv}_{1\times1}\left(F_i\right)\right) \tag{42}$$

where $F_i \in \mathbb{R}^{B\times C\times H\times W}$ is the feature map from a given model, B is the batch size, C is the number of channels, and H and W are the spatial resolution.

Then, we perform global average pooling on the feature maps F_1 and F_2 to obtain the global feature representations Z_1 and Z_2 for each sample. At the same time, to ensure the stability of the feature similarity computation, we perform $L2$ normalization on the global feature vectors. This step is expressed as

$$Z_i = \frac{1}{H\times W}\sum_{h=1}^{H}\sum_{w=1}^{W} F_{\text{global},i}\left(:,:,h,w\right) \tag{43}$$

$$\hat{Z}_i = \frac{Z_i}{\|Z_i\|}. \tag{44}$$

After obtaining the global features, we calculate the similarity between them. We use cosine similarity to measure the degree of similarity between samples. Finally, to optimize the differentiation of global features, we use the cross-entropy loss to define the global contrastive loss as

$$S_{\text{global}} = \hat{Z}_1 \cdot \hat{Z}_2^{\text{T}} \tag{45}$$

$$L_{\text{global}} = \frac{1}{B}\sum_{i=1}^{B} -\log\frac{\exp\left(S_{\text{global}}\left(i,i\right)/\tau\right)}{\sum_{j=1}^{B}\exp\left(S_{\text{global}}\left(i,j\right)/\tau\right)} \tag{46}$$

where $S_{\text{global}} \in \mathbb{R}^{B\times B}$ is the global similarity matrix, which represents the similarity of each sample to other samples, and τ is a temperature parameter to control the smoothing of the similarity distribution. Moreover, $S_{\text{global}}(i,i)$ represents the global feature similarity of the same image in different models (positive sample) and $S_{\text{global}}(i,j)$ denotes the global feature similarity of different similarities between samples (negative samples).

f) Adaptive temperature adjustment and class weighting: Temperature parameter τ is dynamically adjusted as the training epochs increase, allowing tolerance for noise and rapid clustering during early training and converging to a finer discrimination criterion in later stages, which improves the model's capability to distinguish similarities and differences. The specific formula is as follows:

$$\tau_{\text{epoch}} = \tau_0 \times \left(1 - \frac{\text{epoch}}{\text{max_epochs}}\right) \tag{47}$$

where τ is the initial temperature value (0.07), epoch denotes the current number of training rounds, and max_epochs is the maximum number of training rounds.

Additionally, to mitigate the impact of long-tail distribution, a learnable class weight vector $w \in \mathbb{R}^{C\text{num}}$ (where C_{num} is the number of classes) is introduced that adaptively amplifies the contribution of rare classes in the loss calculation, thus improving the recognition of underrepresented classes. The final block-level pixel loss, after weighting by class, is

$$L_{\text{patch}}^{\text{weighted}} = \sum_{c=1}^{C_{\text{num}}} w_c \cdot L_{\text{patch}}^{(c)} \tag{48}$$

where $L_{\text{patch}}^{(c)}$ represents the contrastive loss within category c and w_c denotes the learnable category weights that are used to dynamically adjust the contribution of each category in the loss

Ultimately, the globalcontrastive loss L_{global} is combined with the local pixel contrastive loss $L_{\text{patch}}^{\text{weighted}}$ to form the final RPCL loss

$$L_{\text{RPCL}} = \alpha\left(L_{\text{patch}}^{\text{weighted},1} + L_{\text{patch}}^{\text{weighted},2}\right) + \beta\left(L_{\text{global}}^{1} + L_{\text{global}}^{2}\right) \tag{49}$$

where α and β are hyperparameters for balancing the block-level pixel contrastive loss and the global contrastive loss, respectively.

D. Total Loss

During training, the total loss L_{total} is composed of two parts: the segmentation loss L_{seg} for the semantic segmentation task and the bidirectional CL loss L_{CL} for CL. For the segmentation loss, ground truth labels y supervise the predictions $p_1(\text{out}_1)$ and $p_2(\text{out}_2)$ of FLDNet and SGSNet, and a focal loss is applied to further supervise the prediction maps. This down-weights well-classified samples to address class imbalance. The formulation is as follows:

$$L_{\text{seg}} = \text{FL}\left(y,p\right) = -\sum_{i=1}^{C}\alpha_i y_i\left(1-p_i\right)^{\gamma}\log\left(p_i\right) \tag{50}$$

where y_i denotes the true label for class i, p_i is the predicted probability for class i, C is the number of classes, α_i is a balancing factor for the class weights, γ is the focusing parameter to reduce the weight of easily classified samples, and FL$(\cdot)$ denotes the focal loss.

Finally, the total loss during training is given by

$$L_{\text{total}} = L_{\text{seg}} + L_{\text{CL}} = L_{\text{seg}} + L_{\text{BFR},1\to2} + L_{\text{BFR},2\to1} + L_{\text{RPCL}}. \tag{51}$$

IV. Experiments and Results

A. Datasets and Evaluation Metrics

We evaluated the proposed HCL-Net framework on two benchmark remote sensing datasets: Vaihingen and Potsdam, both annotated with six semantic classes (low vegetation, impervious surfaces, buildings, cars, trees, and clutter/background). The Vaihingen dataset comprises 33 RGB images of size 2500×2000 along with the corresponding nDSM Data. We selected 11 images for training, 17 for testing, and the remaining 5 for validation. The Potsdam dataset contains 38 RGB images of size 6000×6000 with the corresponding nDSM data; in our experiments, 31 images were used for training and testing, and the remaining 7 were used for validation. Because of the large image sizes, all inputs were cropped into 256×256 patches before training. For quantitative evaluation, we report intersection over union (IoU), mIoU, overall accuracy (Acc), and mean accuracy (mAcc).

B. Implementation Details

To verify the effectiveness of the proposed HCL-Net framework, two experimental setups were designed: 1) independent

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TABLE I

QUANTITATIVE COMPARISON OF THE VAIHINGEN DATASET. THE BEST AND SECOND RESULTS ARE HIGHLIGHTED IN RED AND GREEN. ALL BASELINE RESULTS ARE OBTAINED USING ORIGINAL SOURCE CODES UNDER IDENTICAL EXPERIMENTAL CONDITIONS

Models	Imp.surf		Building		Low veg		Tree		Car		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
FCN-8S$_{2015}$[4]	89.66	79.71	93.22	86.80	75.83	64.33	89.22	75.58	45.12	40.16	78.61	69.32
U-Net$_{2015}$[5]	91.68	80.90	89.84	86.50	77.97	65.91	91.30	77.86	75.80	71.22	79.75	71.34
DeepLabv3+$_{2018}$[7]	90.06	81.11	87.04	82.7	76.65	64.44	88.60	76.64	42.51	43.10	76.97	69.49
RedNet$_{2018}$[9]	91.49	84.62	94.81	91.07	78.67	66.59	91.41	78.27	59.77	56.06	83.23	75.32
ACNet$_{2019}$[10]	91.95	85.34	95.45	91.82	78.64	66.87	91.20	78.55	83.12	76.81	88.07	79.88
SA-Gate$_{2020}$[54]	90.99	85.70	93.85	91.72	84.95	68.68	89.06	79.15	84.27	78.07	88.62	80.67
HRCNet$_{2021}$[11]	91.62	81.60	91.72	88.01	79.24	67.38	90.55	78.58	70.69	68.73	84.76	76.86
ESANet$_{2021}$[36]	92.09	85.18	94.93	91.16	75.72	65.48	92.35	77.65	75.92	70.11	86.20	77.92
CMXNet$_{2022}$[37]	91.78	85.39	95.15	91.28	77.00	66.29	92.59	78.85	83.23	78.58	87.95	80.08
MANet$_{2022}$[40]	91.57	83.23	93.77	90.04	76.29	65.59	91.91	77.94	86.03	81.01	87.91	79.56
MGCNet$_{2022}$[38]	91.18	84.89	95.58	91.52	76.22	65.74	92.76	78.53	79.56	75.33	87.06	79.20
SFAF-MA$_{2023}$[41]	90.94	85.37	95.67	92.21	80.99	67.55	90.78	78.95	74.74	71.19	86.62	79.05
EAEFNet$_{2023}$[42]	93.78	84.37	95.35	92.25	79.87	67.98	89.11	79.00	81.54	74.24	87.93	79.57
CEKD$_{2023}$[20]	92.22	85.05	95.78	91.66	76.57	66.77	92.69	78.97	76.63	72.01	86.71	78.89
CATNet$_{2024}$[34]	90.53	83.98	95.27	91.08	78.27	65.62	90.57	77.72	70.28	61.85	84.98	76.05
CAINet$_{2024}$[35]	91.29	84.89	95.65	92.21	78.61	66.85	91.55	78.67	82.60	78.15	87.94	80.15
UNetMamba$_{2025}$[55]	91.98	85.13	95.71	92.40	77.79	67.16	92.13	79.04	84.43	79.20	88.41	80.60
AFENet$_{2025}$[56]	91.88	84.52	93.96	91.02	80.29	67.66	91.03	78.59	82.94	78.92	88.02	80.14
CPSSNet$_{2025}$[57]	93.28	83.28	95.31	90.71	71.28	64.44	94.02	79.04	83.05	77.26	87.39	78.95
FLDNet	92.22	84.15	94.94	91.84	78.72	66.84	90.41	78.30	81.99	76.82	87.66	79.59
SGSNet	92.20	85.99	95.43	91.70	77.45	67.56	93.12	79.22	81.14	75.39	87.87	79.97
FLDNet with CL	92.05	86.07	95.92	92.47	80.09	68.97	91.81	79.79	89.76	83.30	89.92	82.12
SGSNet with CL	92.23	86.62	95.80	92.73	79.84	69.01	92.79	79.91	89.09	82.54	89.95	82.16

TABLE II

QUANTITATIVE COMPARISON OF THE POTSDAM DATASET. THE BEST AND SECOND RESULTS ARE HIGHLIGHTED IN RED AND GREEN. ALL BASELINE RESULTS ARE OBTAINED USING ORIGINAL SOURCE CODES UNDER IDENTICAL EXPERIMENTAL CONDITIONS

Models	Imp.surf		Building		Low veg		Tree		Car		Cluter		mAcc	mIoU
	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU		
FCN-8S$_{2015}$[4]	89.47	79.77	90.69	83.60	85.13	71.12	82.86	71.23	91.02	81.53	49.05	36.49	78.61	69.32
U-Net$_{2015}$[5]	90.03	80.27	88.71	82.92	85.82	71.60	84.06	72.05	93.89	90.24	50.30	36.26	82.13	72.22
DeepLabv3+$_{2018}$[7]	91.57	82.49	91.78	87.59	87.36	73.63	85.45	73.32	93.89	90.04	53.80	43.54	83.97	75.10
RedNet$_{2018}$[9]	92.19	82.83	93.61	90.13	87.00	73.22	83.00	71.77	93.36	90.08	56.74	43.51	84.32	75.26
ACNet$_{2019}$[10]	91.32	82.74	93.83	90.06	86.16	73.53	86.03	72.87	93.79	90.43	54.51	41.65	84.27	75.21
SA-Gate$_{2020}$[54]	85.84	80.64	93.65	88.51	86.46	72.71	85.70	72.89	92.18	89.39	62.70	40.59	84.42	74.12
HRCNet$_{2021}$[11]	90.03	81.68	90.87	85.75	88.17	73.18	82.02	71.32	93.94	89.82	56.72	40.03	83.63	73.63
ESANet$_{2021}$[36]	91.38	82.92	93.69	89.82	87.10	73.16	82.48	70.81	93.08	88.53	55.68	43.38	83.90	74.77
CMXNet$_{2022}$[37]	90.54	81.86	93.80	89.28	87.17	73.11	83.52	71.45	92.23	87.99	51.61	40.58	83.14	74.04
MANet$_{2022}$[40]	90.48	81.20	87.58	83.47	88.78	72.74	82.69	71.54	90.96	88.73	46.16	38.11	81.11	72.63
MGCNet$_{2022}$[38]	90.61	81.21	93.87	89.02	86.72	72.67	83.70	72.51	93.39	88.91	53.24	39.66	83.59	74.00
SFAF-MA$_{2023}$[41]	90.63	82.06	93.31	89.67	85.76	72.35	83.51	70.95	92.55	88.34	60.03	42.40	84.30	74.30
EAEFNet$_{2023}$[42]	92.32	82.04	93.22	89.54	84.34	72.37	85.02	71.91	94.64	88.30	53.55	39.34	83.85	73.92
CEKD$_{2023}$[20]	91.09	80.89	93.37	88.97	84.70	71.71	85.18	71.62	93.01	89.35	52.63	38.18	83.33	73.45
CATNet$_{2024}$[34]	88.12	79.80	93.32	89.52	85.91	70.66	78.78	68.85	91.56	84.99	55.58	40.41	82.21	71.87
CAINet$_{2024}$[35]	88.77	81.25	92.21	88.70	88.50	71.80	81.55	71.17	94.33	89.65	57.05	42.95	83.74	74.25
UNetMamba$_{2025}$[55]	90.67	83.00	94.07	89.53	86.34	73.45	86.31	73.21	93.69	89.61	54.11	42.21	84.20	75.17
AFENet$_{2025}$[56]	91.52	82.18	90.93	86.52	87.26	73.63	83.87	72.97	94.53	90.89	56.84	42.08	84.16	74.71
CPSSNet$_{2025}$[57]	88.87	81.23	91.82	85.61	89.02	73.18	81.38	71.95	93.23	89.56	54.76	38.48	83.18	73.34
FLDNet	90.97	80.59	92.39	88.98	84.74	70.69	81.54	69.76	93.94	87.99	53.81	39.85	82.90	72.98
SGSNet	91.18	82.11	93.69	89.36	87.43	72.74	82.95	71.27	94.20	87.64	44.62	36.83	82.35	73.32
FLDNet with CL	90.28	82.99	93.19	89.78	86.98	73.49	86.40	73.77	94.01	90.36	59.85	43.14	85.12	75.59
SGSNet with CL	91.47	83.13	94.07	90.09	86.90	73.80	84.61	73.26	93.86	90.12	57.89	43.07	84.80	75.58

training, in which FLDNet and SGSNet were trained separately; and 2) collaborative training, in which the BFR and RPCL strategies were used for complementary knowledge transfer between the two models to jointly optimize segmentation performance. All experiments were conducted on a workstation with an NVIDIA TITAN XP GPU (12 GB) using PyTorch. Networks were trained end-to-end for 200 epochs with Adam optimizer (initial learning rate = $1e^{-4}$, weight decay = $5e^{-4}$, momentum = 0.9, and learning rate reduced $\times 0.1$ every 20 epochs). Data augmentation included random scaling, cropping, and rotation.

1) *Baseline Protocols:* For fair comparison, all baseline methods (Tables I and II) were reproduced using the official implementations released by the authors or publicly

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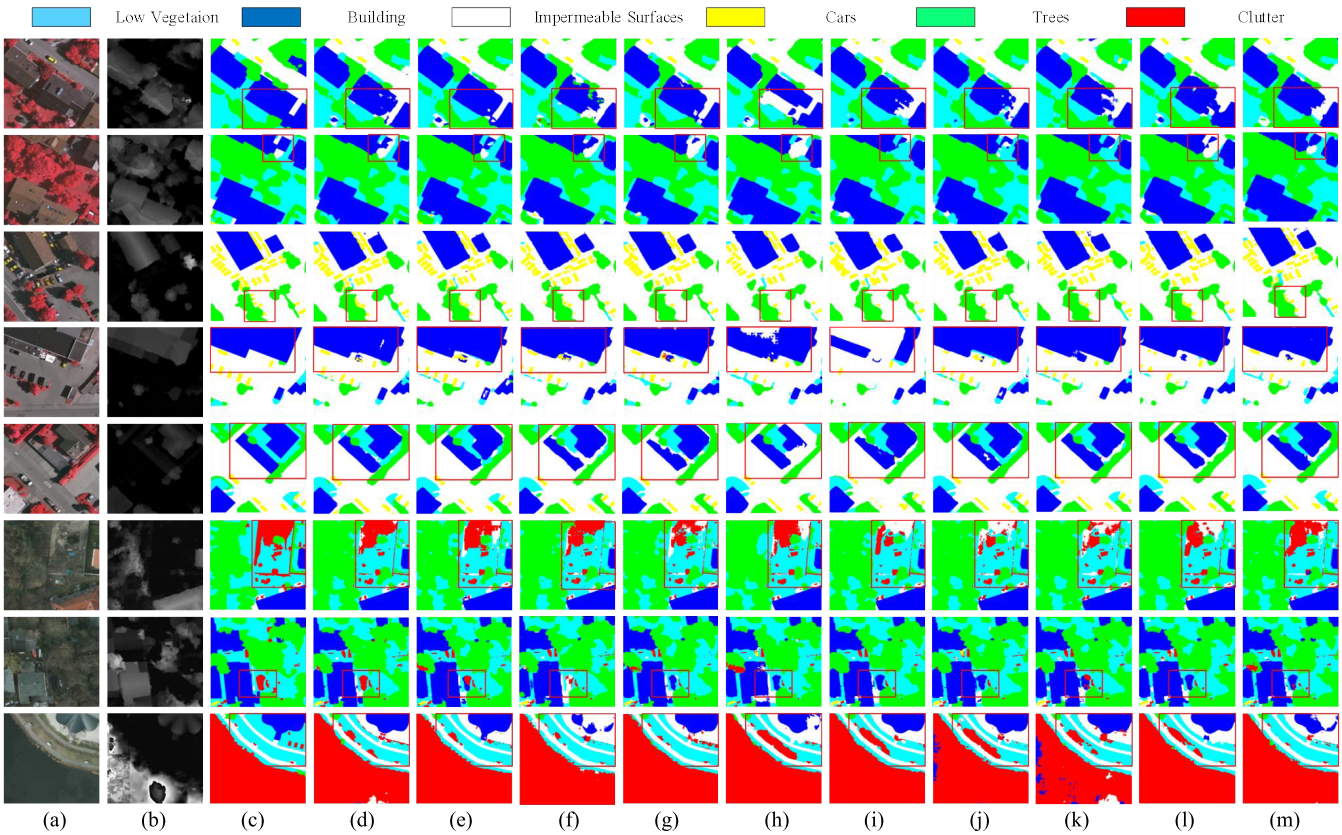

Fig. 4. Semantic segmentation results. (a) RGB, (b) nDSM, (c) GT, (d) FLDNet (with CL), (e) SGSNet (with CL), (f) FLDNet, (g) SGSNet, (h) HRCNet, (i) RedNet, (j) SFAF-MA, (k) CEKD, (l) CMANet, and (m) MGCNet. For ease of observation, we use red boxes to mark salient areas of the predictions.

available repositories when full details were not provided. All experiments were run under identical conditions (same GPU, PyTorch framework, preprocessing pipeline, and evaluation metrics). This ensures that observed performance differences are attributable to algorithmic improvements rather than implementation or hardware variations.

2) *Statistical Validation:* To establish robustness, each experiment was repeated with ten different random seeds (42, 123, 456, 789, 1024, 2021, 3306, 4096, 5555, and 9999). Reported values represent the average of the top ten runs, mitigating the influence of random initialization and highlighting consistent performance gains. This procedure confirms that improvements arise from the proposed algorithm rather than stochastic variance.

C. Performance Comparison and Result Analysis

1) Quantitative Analysis: Tables I and II report the results obtained for FLDNet and SGSNet when these models were trained independently and after collaborative training. Both models exhibited significant improvements following the bidirectional complementary knowledge transfer. On the Vaihingen dataset, collaborative training improved the FLDNet mAcc and mIoU scores by approximately 2.26% and 2.53%, respectively, and those of SGSNet by approximately 2.08% and 2.19%, respectively, compared with the scores for independent training. Similar improvements were observed on the Potsdam dataset. Compared with various state-of-the-art (SOTA) methods, HCL-Net achieved higher mAcc and mIoU on both datasets, demonstrating its superiority in handling complex remote sensing scenes with large intraclass variations and high interclass similarities.

2) Qualitative Analysis: To visually verify the segmentation performance, segmentation maps from comparative experiments were considered (Fig. 4). The models trained using the collaborative strategy extracted more precise edge information for complex objects, particularly for fine-grained targets such as buildings and vehicles. Compared with other methods, the proposed method exhibited less segmentation blurring due to illumination changes and texture complexity, yielding more compact intraclass features and clearer interclass distinctions.

3) Complexity Comparison: In practice, a lightweight model and computational efficiency are crucial. As reported in Table III, FLDNet and SGSNet had parameter counts of only 24.25 and 12.36 M, respectively, with computational costs of 6.13 and 6.35 G. More importantly, both models demonstrate superior inference speed with FPS of 28.4 and 32.1, significantly outperforming most SOTA methods while maintaining high segmentation accuracy. This combination of low complexity and high inference speed makes HCL-Net highly suitable for deployment in resource-constrained environments.

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TABLE III
COMPUTATIONAL COMPLEXITY COMPARISON

Models	Backbone	Params(M)↓	Flops(G)↓	FPS↑
U-Net$_{2015}$[5]	VGG-16	26.36	55.93	12.3
FCN-8S$_{2015}$[4]	VGG-16	134.29	74.55	8.7
RedNet$_{2018}$[9]	ResNet-50	81.95	21.17	18.5
ACNet$_{2019}$[10]	ResNet-50	116.6	26.41	15.2
SA-Gate$_{2020}$[54]	ResNet-101	110.85	41.23	11.8
HRCNet$_{2021}$[11]	ResNet-50	62.71	30.27	14.6
ESANet$_{2021}$[36]	ResNet-34	45.42	10.15	22.1
MANet$_{2022}$[40]	ResNet-50	35.86	19.49	19.3
CMXNet$_{2022}$[37]	SegFormer	66.56	14.26	20.8
MGCNet$_{2022}$[38]	ConvNeXt	209.99	47.51	9.4
SFAF-MA$_{2023}$[41]	ResNet-50	232.08	18.24	16.7
EAEFNet$_{2023}$[42]	ResNet-50	77.97	23.3	17.9
CEKD$_{2023}$[20]	ResNet-50	245.03	68.61	7.2
CATNet$_{2024}$[34]	Swim-Former	262.74	76.79	6.8
CAINet$_{2024}$[35]	MobileNet-V2	13.8	4.69	35.7
UNetMamba$_{2025}$[55]	rest-lite	14.76	3.88	21.2
FLDNet	ResNet-18	24.25	**6.13**	28.4
SGSNet	DFormer-T	**12.36**	6.35	**32.1**

TABLE IV
MODULE ABLATION EXPERIMENTS

Modules	VAIHINGEN		POTSDAM	
	mAcc	mIoU	mAcc	mIoU
Module Ablations				
FLDNet w/o WPAFB	86.00	77.37	81.73	71.91
SGSNet w/o DTEP	86.25	76.21	81.67	71.69
Individual Models				
FLDNet	87.66	79.59	82.90	72.98
SGSNet	87.87	79.97	82.35	73.32
Simple Cross-Model Transfer Baselines				
FLDNet + Simple KD (L2)	86.23	77.95	82.15	72.34
SGSNet + Simple KD (L2)	86.41	78.12	82.28	72.48
FLDNet + Feature Mimicking	86.35	78.03	82.07	72.21
FLDNet + Feature Mimicking	86.52	78.24	82.33	72.55
Progressive Collaborative Training				
FLDNet+ BFR	88.60	80.99	83.54	74.34
SGSNet+ BFR	88.47	81.20	83.34	74.02
FLDNet+RPCL	88.86	81.04	83.91	74.90
SGSNet+RPCL	88.68	81.04	83.53	74.07
FLDNet+BFR+RPCL	89.92	82.12	85.12	75.59
SGSNet+BFR+RPCL	89.95	82.16	84.80	75.58

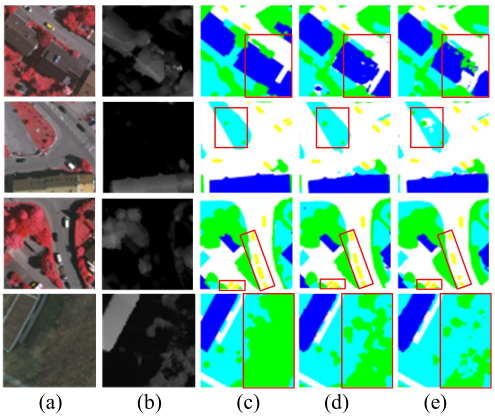

Fig. 5. Visual comparison of FLDNet with and without WPAFB module. (a) RGB, (b) nDSM, (c) GT, (d) FLDNet (with WPAFB and CL), and (e) without WPAFB. Examples include building-dense areas, vegetation-complex regions, and vehicle-dense areas to demonstrate frequency-domain local detail processing capabilities and boundary precision improvements.

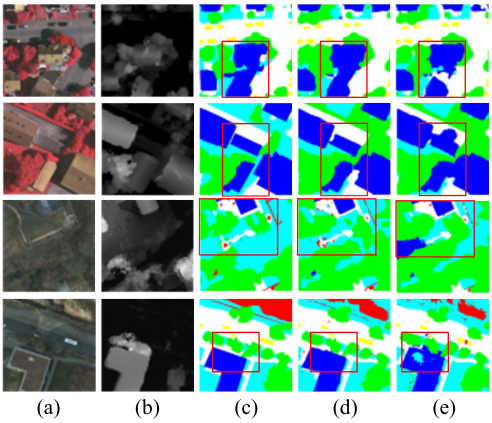

Fig. 6. Visual comparison of FLDNet with and without DTEP module. (a) RGB, (b) nDSM, (c) GT, (d) SGSNet (with DTEP and CL), and (e) without DTEP. Examples showcase spatial-domain global structure perception in various scene complexities with emphasis on edge extraction and texture detail preservation.

D. Ablation Studies

To evaluate the effectiveness of the proposed network modules and CL strategies, several ablation experiments were conducted on the Vaihingen and Potsdam datasets. All ablation experiments used the same training strategy and data augmentation as the complete model for fair comparison. Detailed results are provided in Table IV.

1) WPAFB Module: In FLDNet, the WPAFB module is used to achieve adaptive fusion of low- and high-frequency information from the RGB and nDSM modalities in the frequency domain. As shown in Fig. 5, the visual analysis across building-dense, vegetation-complex, and vehicle-dense scenarios demonstrates WPAFB's effectiveness in preserving architectural boundaries, distinguishing vegetation types, and detecting small targets through phase–amplitude fusion. Replacing WPAFB with simple addition results in mAcc and mIoU drops of 1.66% and 2.22% on Vaihingen and 1.17% and 1.07% on Potsdam, confirming its key role in detail preservation and illumination interference suppression.

2) DTEP Module: SGSNet relies on the DTEP module to strengthen global context, edge, and texture feature expression through DTEP. Fig. 6 demonstrates DTEP's effectiveness across diverse scenarios, showing improved building footprint completeness, enhanced vegetation classification through multidirectional texture features, and better vehicle detection via positional encoding and edge attention. Replacing DTEP with simple addition results in mAcc and mIoU drops of 1.62% and 3.76% on Vaihingen and 0.68% and 1.63% on Potsdam, validating its importance for global structure perception and edge detail processing.

To validate the necessity of our sophisticated CL strategy, we evaluated simple cross-model knowledge transfer approaches as baselines. Simple KD using $L2$ loss between corresponding features achieved 86.23%/86.41% mAcc for

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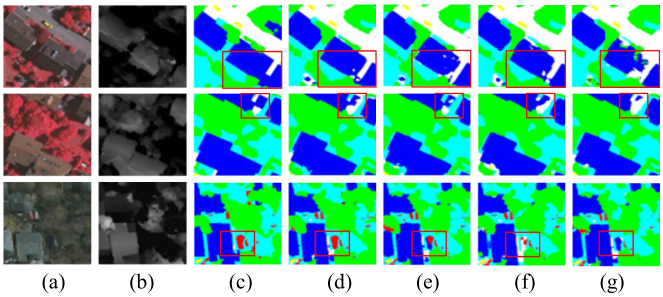

Fig. 7. Visual comparison of FLDNet with and without BFR. (a) RGB, (b) nDSM, (c) GT, (d) FLDNet (with CL), (e) without BFR (FLDNet), (f) SGSNet (with CL), and (g) without BFR (SGSNet).

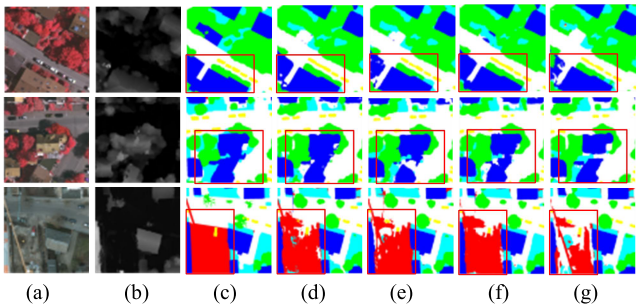

Fig. 8. Visual comparison of FLDNet with and without RPCL. (a) RGB, (b) nDSM, (c) GT, (d) FLDNet (with CL), (e) without RPCL (FLDNet), (f) SGSNet (with CL), and (g) without RPCL (SGSNet).

FLDNet/SGSNet on Vaihingen, while feature mimicking approaches showed similar performance (86.35%/86.52% mAcc). Although these simple transfer methods outperform individual models without cross-modal collaboration, they significantly underperform our BFR mechanism (88.60%/88.47% mAcc), demonstrating that direct feature alignment cannot effectively bridge the domain discrepancies between CNN-based and Transformer-based architectures. This validates the necessity of our sophisticated Gram matrix alignment and VID approach for effective heterogeneous CL.

3) BFR Module: The BFR module aligns high-order feature statistics using the Gram matrix and constrains feature distribution via VID to achieve effective cross-model knowledge transfer. As shown in Fig. 7, removing the BFR module and relying solely on RPCL for CL leads to a significant drop in robustness when handling large intraclass variations and high interclass similarities. Table IV reveals that on the Vaihingen dataset, FLDNet's mAcc and mIoU drop by approximately 1.06% and 1.08%, respectively, and SGSNet's metrics drop by about 1.27% and 1.12%. Similar decreases are observed on the Potsdam dataset, confirming the necessity of BFR for mitigating domain discrepancies.

4) RPCL Module: The RPCL module builds contrastive loss at the pixel level through hierarchical sampling and adaptive temperature adjustment, enhancing intraclass feature compactness and suppressing confusion. As illustrated in Fig. 8, removing RPCL leads to significant declines in segmentation accuracy for underrepresented classes (e.g., vehicles or low vegetation) and blurred boundaries between different classes. The quantitative analysis in Table IV indicates that on the Vaihingen dataset, FLDNet's mAcc and mIoU drop by about 1.32% and 1.13%, respectively, and those of SGSNet drop by about 1.48% and 0.96%; again, similar trends are observed on the Potsdam dataset. This verifies the critical role of RPCL in mitigating class imbalance and improving fine-grained segmentation.

The results in Table IV and the corresponding segmentation maps show that the complete model effectively distinguishes buildings, vehicles, and vegetation, whereas removing any module degrades local detail, edge precision, or class discrimination. These ablation studies confirm the critical contribution of each component to HCL-Net's overall performance and establish a solid foundation for further optimization.

TABLE V
QUANTITATIVE EVALUATION OF MODELS ON THE RGB-D MIRROR DATASET

Methods	IOU↑	BER↓	MAE↓
CAINet	74.93	10.42	0.047
MirrorNet	68.37	8.66	0.062
PDNet	77.77	7.77	0.042
SANet	78.43	8.16	0.041
SATNet	78.42	9.31	0.031
FLDNet	78.46	8.10	0.041
SGSNet	78.50	8.05	0.040

E. Cross-Domain Validation on RGB-D Mirror Dataset

To further assess the generalization and robustness of HCL-Net in different application scenarios, we conducted cross-domain experiments on the RGB-D Mirror dataset, which poses a distinct challenge compared with remote sensing segmentation. The dataset contains 3049 RGB-D image pairs (2000 for training and 1049 for testing) targeting mirror surface detection in indoor environments. We adapted HCL-Net to the binary segmentation task and evaluated performance using MAE and BER as primary metrics, with IoU and F-measure for comprehensive assessment. As reported in Table V, both FLDNet and SGSNet retained their CL advantages in this domain, with collaborative training consistently outperforming independent training. SGSNet achieved the best results across all metrics, confirming the transferability of our heterogeneous CL framework to diverse computer vision tasks beyond remote sensing.

V. CONCLUSION

We introduced HCL-Net, a heterogeneous CL framework that integrates FLDNet (ResNet18 + WPAFB for multifrequency detail extraction) and SGSNet (DFormer-T + DTEP for global context modeling). Through BFR and RPCL, HCL-Net enables effective CNN–Transformer collaboration while remaining lightweight. Experiments on the Vaihingen and Potsdam datasets showed clear gains over independent training (+2.26%/+2.08% mAcc and +2.53%/+2.19% mIoU for FLDNet/SGSNet) and superiority to 19 SOTA methods, with only 24.25/12.36 M parameters and 6.13/6.35 G FLOPs. Cross-domain experiments on the RGB-D Mirror dataset further validated its transferability.

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Despite strong results, limitations remain: 1) evaluation focused on European datasets with similar conditions; 2) manual hyperparameter tuning for each dataset; and 3) reliance on CNN–Transformer pairs. Future work will explore adaptive collaboration strategies, diverse modalities (hyperspectral, synthetic aperture radar (SAR), and light detection and ranging (LiDAR)), cross-geographical validation, real-time deployment, and theoretical foundations of heterogeneous CL.

In summary, HCL-Net establishes a lightweight yet accurate paradigm for semantic segmentation, bridging CNN–Transformer tradeoffs and opening pathways for robust, cross-domain applications.

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